

C-R Equations معادلات ريمان

$$f(z) = u(x, y) + i v(x, y)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$f(z) = h(z)$$

Analytic function الدالة تحليلية

$$f(z) = u(x, y) + i v(x, y)$$

(1)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

(2) $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}$

are continuous

$f(z)$ Analytic
in a domain
في مجال

$$f(z) = \frac{1}{z-1}$$

$f(z)$ analytic in $C \setminus \{1\}$

Entire function الدالة الكليّة

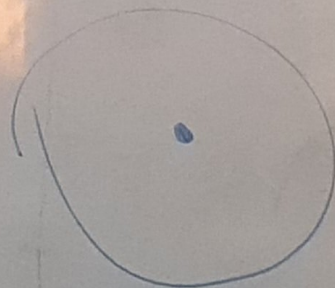
$f(z)$ Entire

If $f(z)$ Analytic everywhere
تحليلية في كل مكان

$$f(z) = z^2 + 3z + 1$$

$$f(z) = e^z$$

Singular point النقطة الشاذة



complex integration

$$\begin{aligned} & \int_0^{\frac{\pi}{3}} \frac{it}{e} dt \\ &= \frac{it}{i} \Big|_0^{\frac{\pi}{3}} \\ &= -i \left(\frac{it}{e} \right) \Big|_0^{\frac{\pi}{3}} \\ &= -i \left(e^{\frac{i\pi}{3}} - e^{0i} \right) \end{aligned}$$

$$\frac{1}{i} = -i$$

$$= -i \left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right) - 1 \right]$$

$$= -i \left[\frac{1}{2} + i\frac{\sqrt{3}}{2} - 1 \right]$$

$$= -i \left[-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right]$$

$$= \frac{i}{2} + -\frac{\sqrt{3}}{2} i^2$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2} i$$

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$e^{-i\theta} = \cos(\theta) - i\sin(\theta)$$

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

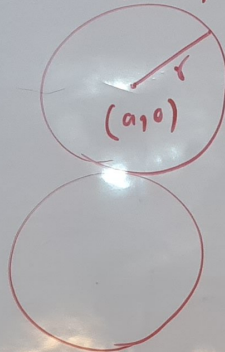
$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

complex integration

$$\oint_C \frac{1}{z-a} dz$$

where C circle $|z-a|=r$
(a, 0)



$$z-a = r e^{i\theta}$$

$$dz = r e^{i\theta} i d\theta$$

$$= \int_0^{2\pi} \frac{1}{r e^{i\theta}} r e^{i\theta} i d\theta$$

$$= \int_0^{2\pi} i d\theta = i \int_0^{2\pi} 1 d\theta = i \theta \Big|_0^{2\pi} = i(2\pi - 0) = 2\pi i$$

$$|x|=1$$
$$x=\pm 1$$

$$|-1|=1$$

$$|1|=1$$

$$|z|=1$$

$$|x+iy|=1$$

$$\sqrt{x^2+y^2}=1$$

$$x^2+y^2=1$$

$$(0,0) \quad r=1$$

$$|z-3|=2$$

$$(3,0) \quad r=2$$

$$|z-2i|=5$$

$$(0,2) \quad r=5$$

complex integration

$n \neq -1$

$$\oint_C (z-a)^n dz$$

where C circle $|z-a|=r$
(a, 0)

$$z-a = r e^{i\theta}$$

$$dz = r e^{i\theta} i d\theta$$

$$\int_0^{2\pi} (r e^{i\theta})^n r e^{i\theta} i d\theta$$

$$\int_0^{2\pi} r^n e^{in\theta} \cdot r e^{i\theta} i d\theta$$

$$0 \leq \theta < 2\pi$$

$$i \int_0^{2\pi} r^{n+1} e^{in\theta + i\theta} d\theta$$

$$i r^{n+1} \int_0^{2\pi} e^{(n+1)\theta i} d\theta$$

$$i r^{n+1} \left[\frac{e^{(n+1)\theta i}}{(n+1)i} \right]_0^{2\pi}$$

$$= \frac{r^{n+1}}{n+1} \left[\frac{e^{(n+1)2\pi i} - e^0}{i} \right]$$

$$= \frac{r^{n+1}}{n+1} \left[\cos((n+1)2\pi) + i \sin((n+1)2\pi) - 1 \right]$$

$$= \frac{r^{n+1}}{n+1} [1 + 0 - 1] = 0$$

$(0, 0)$ $(1, 2)$

$$\frac{y-0}{x-0} = \frac{2-0}{1-0}$$

$$\frac{y}{x} \neq \frac{2}{1}$$

$$y = 2x$$

$$x = 0$$

$(0, 0)$ $(0, 2)$

$$\frac{y-0}{x-0} = \frac{2-0}{0-0}$$

$$\frac{y}{x} \neq \frac{2}{0} \Rightarrow 2x = 0$$

$$x = 0$$

$(1, 2)$ $(1, 3)$

$$x = 1$$

$$\frac{y-2}{x-1} = \frac{3-2}{1-1}$$

$$\frac{y-2}{x-1} \neq \frac{1}{0} \Rightarrow x-1 = 0$$

$$x = 1$$

$(0, 2)$ $(3, 2)$

$$y = 2$$

$$\int_C z^2 dz$$

C the straight line

$$z=0$$

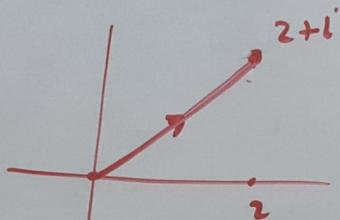
$$z=2+i$$

$$(0, 0) \text{ --- } (2, 1)$$

$$\frac{y-0}{x-0} = \frac{1-0}{2-0}$$

$$\frac{y}{x} = \frac{1}{2}$$

$$2y = x$$



$$s.t. \\ 2+i$$

$$\int_0^{2+i} z^2 dz$$

$$z = x + iy$$

$$z = 2y + iy = (2+i)y$$

$$dz = (2+i)(1) dy$$

$$= \int_0^1 ((2+i)y)^2 (2+i) dy$$

$$= \int_0^1 (2+i)^2 y^2 (2+i) dy$$

$$= (2+i)^2 (2+i) \int_0^1 y^2 dy$$

$$= (4+4i-1)(2+i) \left[\frac{y^3}{3} \right]_0^1$$

$$= (3+4i)(2+i) \left[\frac{1}{3} - 0 \right]$$

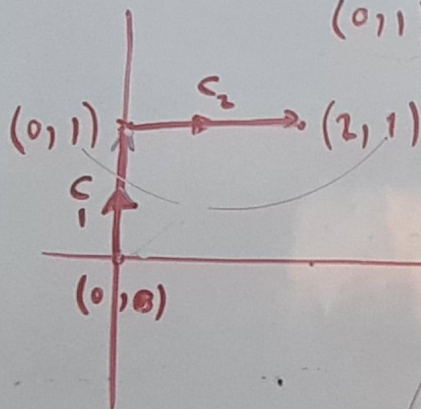
$$= (6+3i+8i-4) \frac{1}{3}$$

$$= \frac{1}{3} (2 + 11i)$$

$$\int_C z^2 dz$$

C the straight line

$z=0$ to $z=i$ $(0,0)$ to $(0,1)$
and ... $z=i$ to $z=2+i$ $(0,1)$ to $(2,1)$



sol

$$= \int_{C_1} z^2 dz + \int_{C_2} z^2 dz$$

$$\int_{C_1} z^2 dz$$

$x=0$

$z=x+iy$

$z=0+iy$

$z=iy$

$dz=i dy$

$$= \int_0^1 (iy)^2 (i dy) = i \int_0^1 y^2 dy = -i \left[\frac{y^3}{3} \right]_0^1 = -i \left(\frac{1}{3} - 0 \right) = -\frac{i}{3}$$

$$\int_{C_2} z^2 dz$$

$y=1$

$z=x+iy$

$z=x+i$

$dz=1 dx$

$$= \int_0^2 (x+i)^2 dx$$

$$= \int_0^2 (x^2 + 2ix - 1) dx$$

$$= \left[\frac{x^3}{3} + 2ix^2 - x \right]_0^2$$

$$= \left(\frac{8}{3} + 4i - 2 \right) - (0 + 0 + 0)$$

$$= \frac{8-6}{3} + 4i = \frac{2}{3} + 4i$$

$$\begin{aligned} & \int_{C_1} z^2 dz \\ &= \int_0^1 (iy)^2 (i dy) \\ &= i \int_0^1 y^2 dy \\ &= -i \left[\frac{y^3}{3} \right]_0^1 \\ &= -\frac{i}{3} \end{aligned}$$